Report: Medical Insurance Charges Analysis

Date: March 31, 2026

To: Medical aid scheme in South Africa.

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Executive Summary

The goal of this project was to develop a sliding scale of insurance charges based on individual member profiles. We utilized a US-based dataset as proof of concept to demonstrate how lifestyle and demographic factors influence costs. The dataset included:

- Demographics: Age, Sex, and Number of Children.
- Health/Lifestyle: BMI and Smoking Status.
- Geography: Residential Region.

Data Understanding & Cleaning

Column	Type	Unique Values
age	int64	18-64
sex	object	female, male
bmi	float64	15.96 - 53.13
children	int64	0-5
smoker	object	Yes, no
region	object	southwest, southeast, northwest, northeast
charges	float64	1,121.87 - 63,770.43

Initial profiling confirmed that all variables fell within expected ranges, with no missing values or obvious inconsistencies, a critical first step before any modeling (Baesens et. al. 2024).

Exploratory Data Analysis (EDA)

Univariate Analysis - Numerical Variables

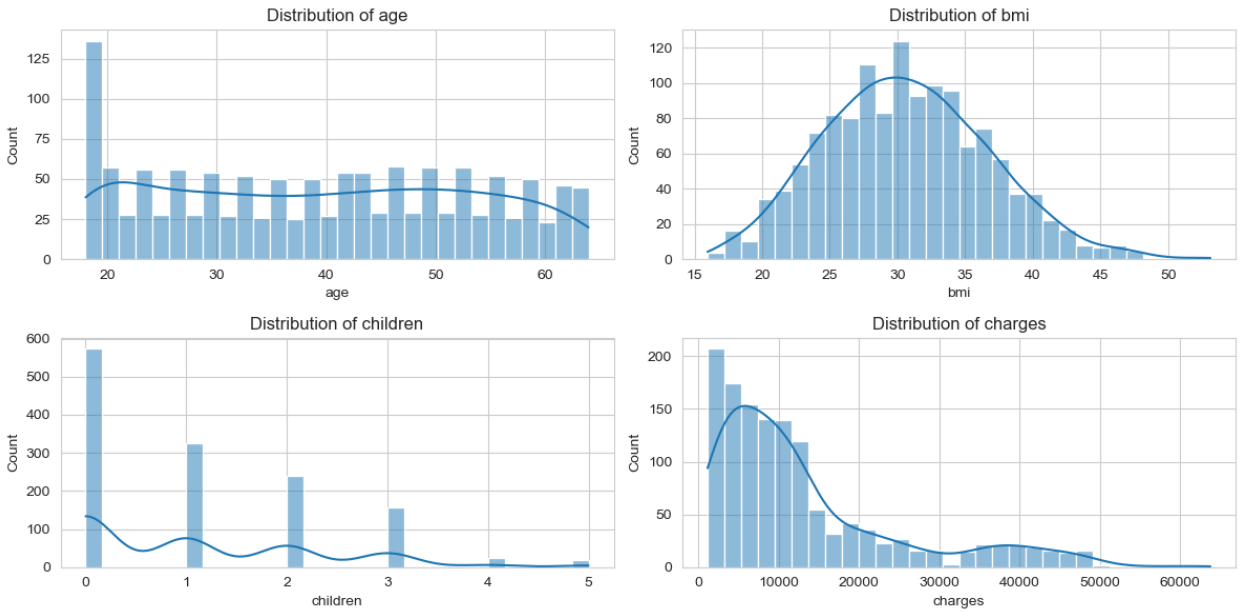

Figure 1: Distribution of Insurance Charges, Age, BMI, and Children.

Insurance Charges

The ***charges distribution*** shows strong right skewness - most patients have charges between $2,000-$10,000, with a long tail reaching up to $63,770. This indicates that a small subset of patients incurs disproportionately high medical costs.

Age, BMI, and Children.

- Age: Roughly uniform distribution (18-64 years)
- BMI: Approximately normal, centered around 30
- Children: Skewed toward 0-3 children, few with 4-5

Univariate Analysis – Categorical Variables

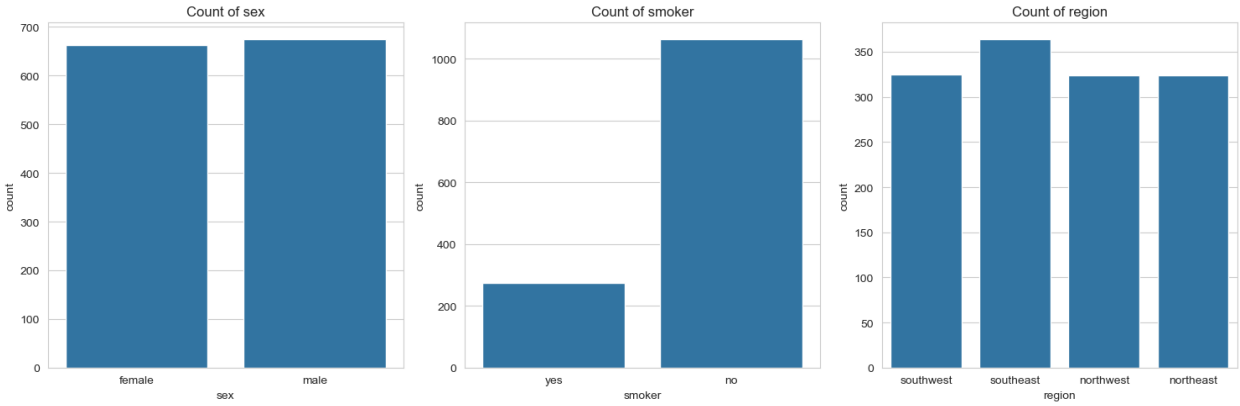

Figure 2: Count Plots for Sex, Smoker, and Region

- Slightly more males (51%) than females (49%)
- Approximately 25% of individuals are smokers
- Region distribution is balanced across all four regions

Bivariate Analysis – Key Relationships

Charges vs. Age

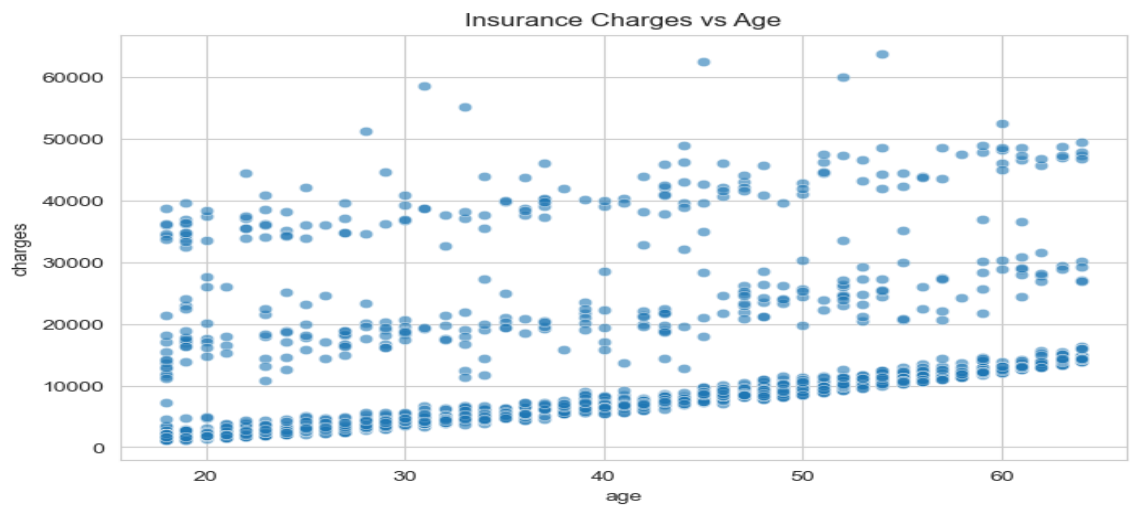

Figure 3: Scatter Plot - Charges vs. Age

- Weak positive correlation (r ≈ 0.30)
- Charges generally increase with age, but variation is substantial
- A distinct cluster of very high charges appears at all ages - these are primarily smokers

Charges vs. BMI

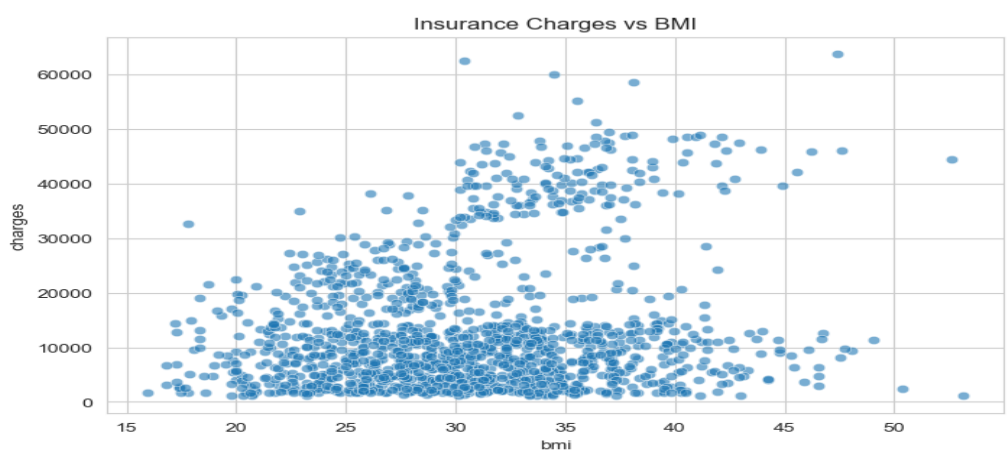

Figure 4: Scatter Plot - Charges vs. BMI

- No strong linear relationship (r ≈ 0.20)
- Highest charges cluster at moderate BMI values (30-40)
- The separation is driven by smoking status

Charges vs. Smoking Status

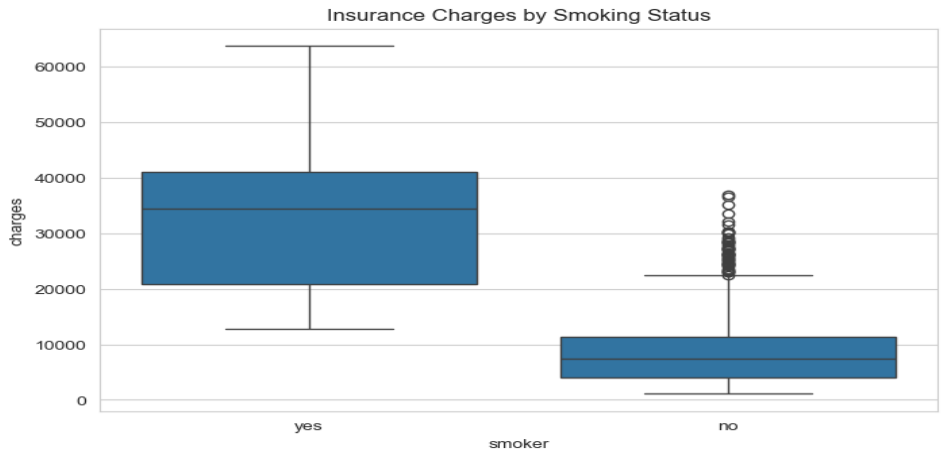

Figure 5: Box Plot - Charges by Smoking Status

Statistic	Non-Smoker	Smoker
Median	~$7,000	~$34,000
IQR	$4,000-$10,000	$27,000-$40,000
Max	~$35,000	~$63,770

Insight: This is the strongest single factor - smokers pay dramatically higher premiums.

Charges vs. Sex & Region

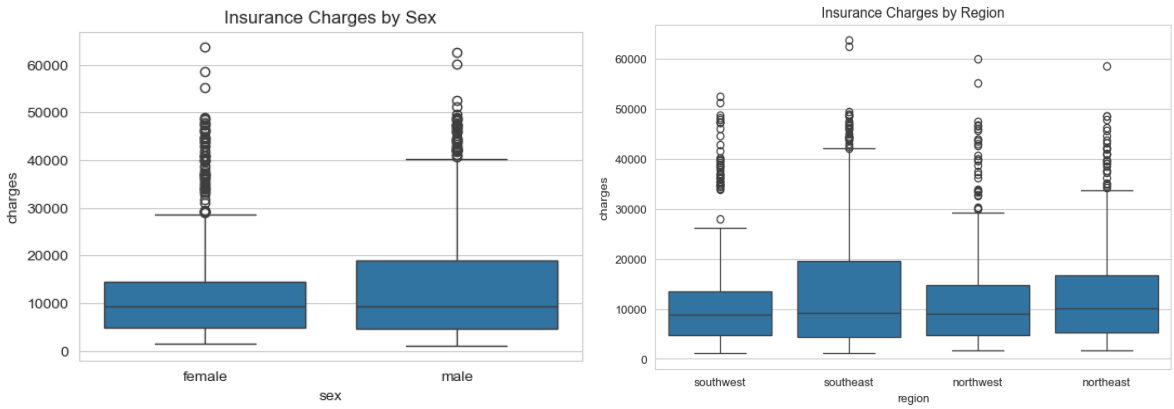

Figure 6: Box Plots - Charges by Sex and by Region

- Sex shows minimal difference in median charges
- Region shows slight variation (Southeast slightly higher), but not statistically significant

Multivariate Analysis

Correlation Matrix

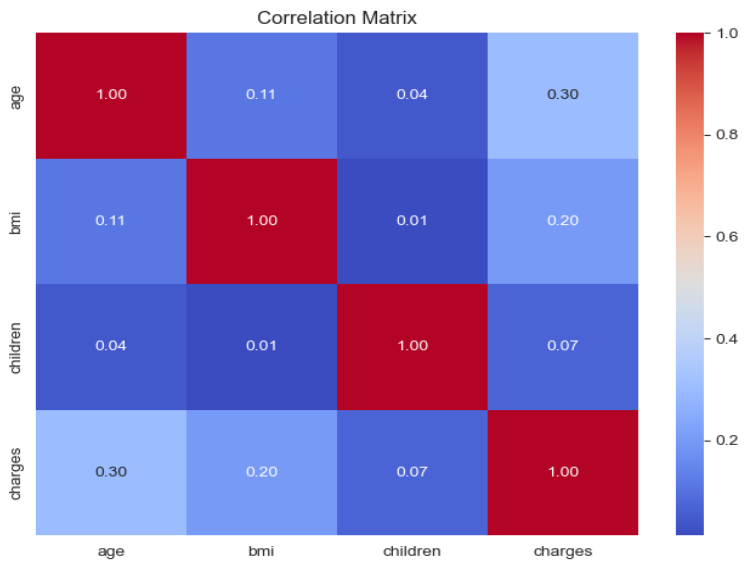

Figure 7: Correlation Heatmap (age, bmi, children, charges)

Variable	Correlation with Charges
age	0.30
bmi	0.20
children	0.07

Insight: Age has a weak positive correlation with charges (0.30). BMI and children have very weak correlations. This suggests that charges are not linearly related to these variables alone; smoking likely interacts with them.

Pairplot Colored by Smoker

Figure 8: Pairplot (age, bmi, children, charges)

This visualization reveals the key pattern:

- Non-smokers (orange): Low charges regardless of age/BMI
- Smokers (blue): Charges increase with age and BMI
- Clear separation between the two groups

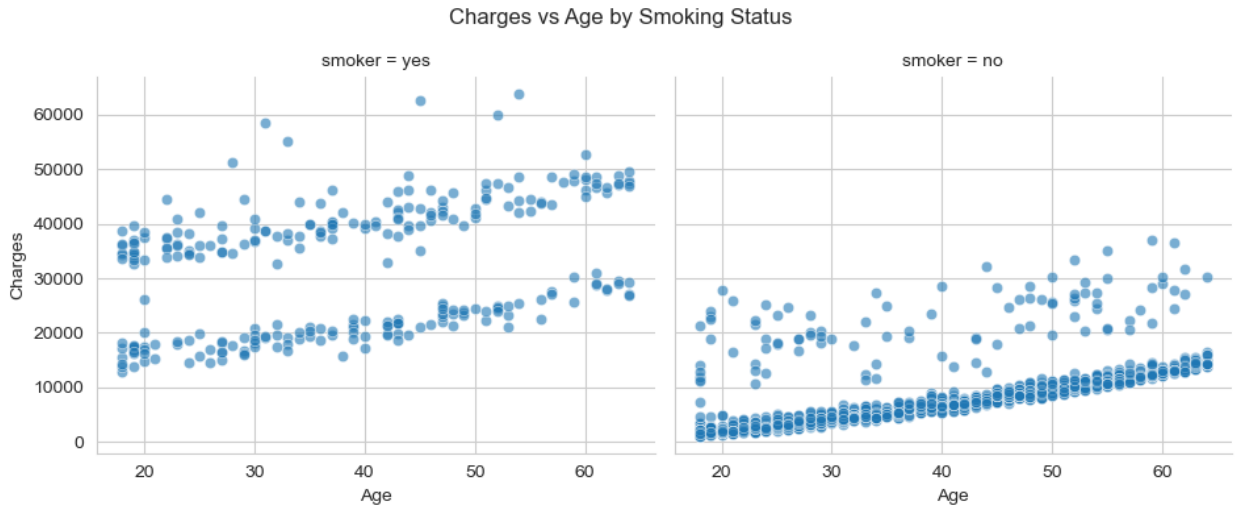

Figure 9: Faceted Scatter Plots - Charges vs. Age by Smoking Status

Insight: Among smokers, charges increase with age; among non-smokers, charges are consistently low across all ages.

Data Preparation & Engineering

To translate real-world demographics into a format the model can process, we performed the following:

- Categorical Encoding: sex, smoker status, and region were converted into numerical "dummy" variables.
- Feature Selection: Using an F-test, identified the five most impactful drivers of cost:
    - Top 5: Age, BMI, Number of Children, Smoking Status, and Region (Southeast).

- Target Variable (y): The model specifically predicts the total insurance charges.

Correctly encoding categorical variables and choosing the most important features are essential steps that directly affect how well a model works and how easy it is to understand (Baesens et. al. 2024).

Model Development & Evolution

Progressed from a simple linear approach to a model to better capture complex lifestyle interactions.

- Initial Model: Simple Linear Regression
    - Baseline Performance: Explained 78.2% of the variance in costs.
    - Weakness: The Residual Plot revealed heteroscedasticity (error spread increases as charges go up), meaning the model struggled with high-cost outliers.

Linear regression usually assumes that the errors have a constant variance; however, when this assumption is not met (known as heteroscedasticity), it becomes necessary to use more adaptable models or to transform the features (Secondary Analysis of Electronic Health Records, 2016).

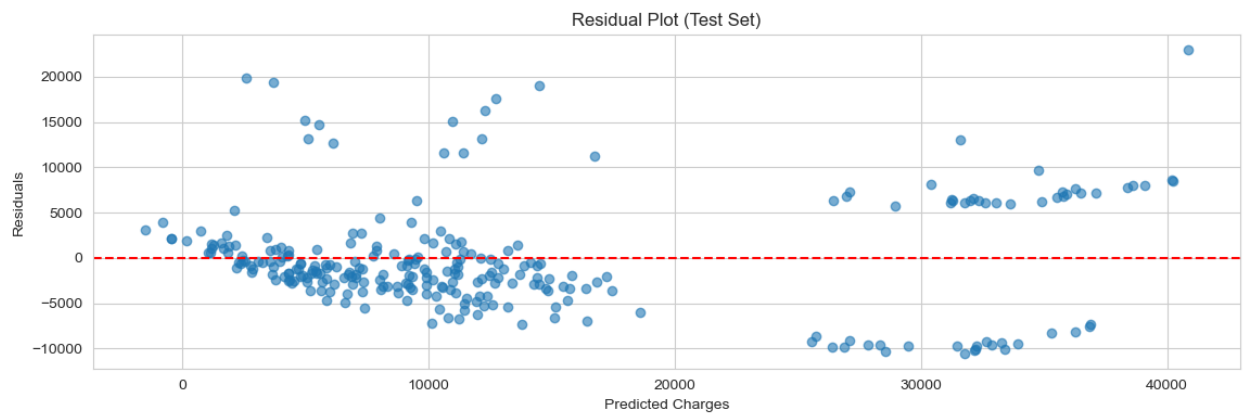

Figure 10: Probability Plot of Prediction Errors

- Final Model: Polynomial Regression (Degree 2)
    - By introducing interaction terms, we significantly increase accuracy.
    - Explained Variance (R^2): Increased to 87%.
    - Error Reduction: The Mean Absolute Error (MAE) dropped by approximately $1,500, meaning predictions are significantly closer to actual costs.

<u>Retrained Model (Polynomial Features, degree=2)</u>

Train R^2: 0.837

Test R^2: 0.870

Mean Absolute Error: $2733.41

Root Mean Squared Error: $4497.11

<u>Business Interpretation</u>

- Test R^2 - 87% of cost fluctuations are successfully explained.

- MAE - On average, predictions are off by this amount.
- Train ≈ Test - The model performs consistently on new, unseen data.

Polynomial and interaction features help models understand non-linear relationships, which is a well-known method in health economics and actuarial modeling (Secondary Analysis of Electronic Health Records, 2016).

South African Deployment Roadmap

The proof of concept is ready for local adaptation. To deploy this for a South African medical aid scheme:

- Localize Geography: Swap the four US regions for South African provinces.
- Retrain on Local Data: Utilize the existing Polynomial Features pipeline on South African member data.
- Refine Factors: Integrate local drivers such as urban vs. rural status and proximity to public vs. private hospital networks.
- Sliding Scale: The model's coefficients provide the "sliding scale" requested, allowing for transparent premium adjustments based on individual risk profiles.

Training the model again using local data and adding variables specific to the region helps keep it relevant and applicable. Which is an important idea in the secondary analysis of electronic health records (Secondary Analysis of Electronic Health Records, 2016).

Bibliography

Baesens, B., Verbeke, W., De Smedt, J., De Weerdt, J. and Weytjens, H. (n.d.). AI FOR BUSINESS. Data Preprocessing and Feature Engineering. [pdf] Available at: < https://www.bluecourses.com/asset-v1:bluecourses+BC3+October2019+type@asset+block/BartBaesensDataPreprocessingFeatureEng.pdf > [Accessed 15 March 2026].

Secondary Analysis of Electronic Health Records. (2016). [online] PubMed. Cham (CH): Springer. Available at: < https://www.ncbi.nlm.nih.gov/books/NBK543630/ > [Accessed 15 March 2026].